

Lab#04

Department of Computer Science (7th Semester)

Course: DevOps

Name:

Zahra Bashir

SUBMITTED TO:

Sir Waqas Saleem

Registration Number

23-BCS-092(Section-A)

DevOps Repository Link: https://github.com/zahra225/DevOps

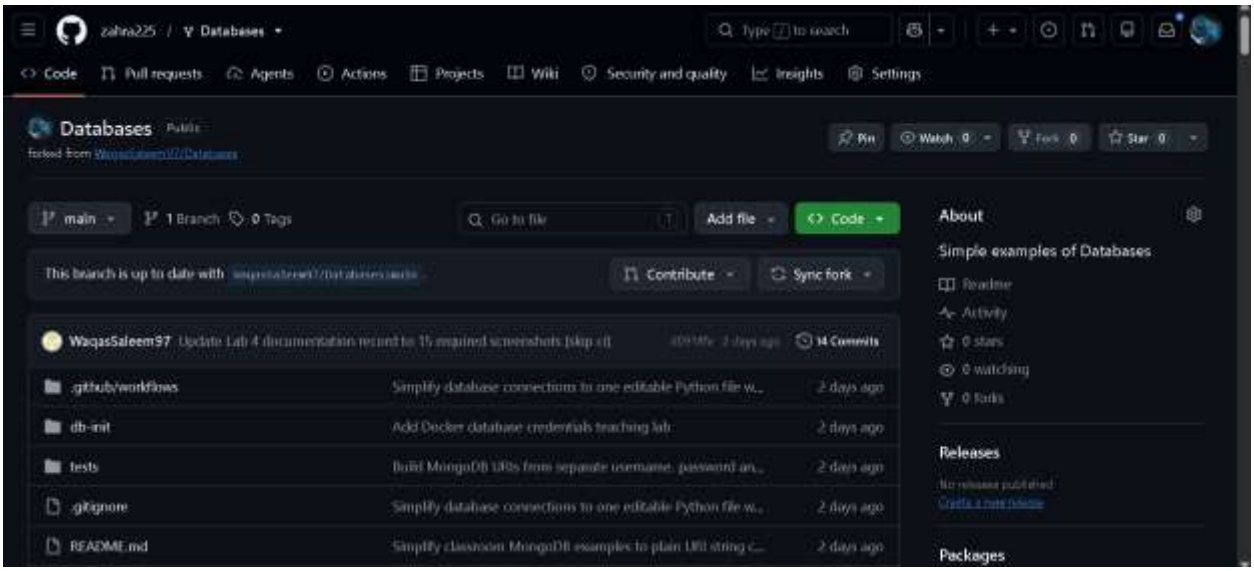
Databases Fork Repository Link: https://github.com/zahra225/Databases

Lab Title: Working with Git History, Stashing, and Reverting Commits

Github link: [zahra225]

Task 1: Fork and Clone the Databases Repository

- **databases_fork.png**

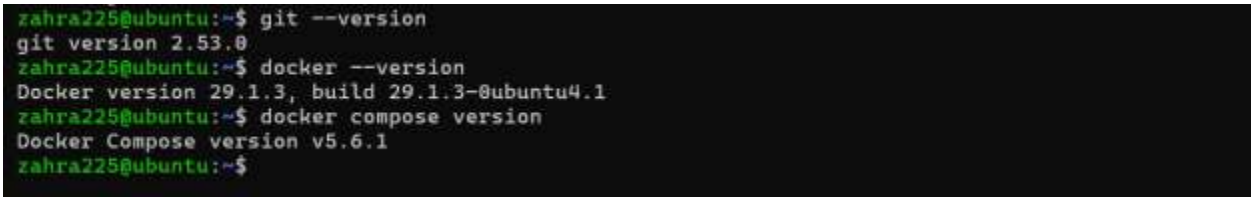


- **databases_clone.png**



Task 2: Install Docker, Docker Compose, and Python on Ubuntu

- **docker_installation.png**



Task 3: Start the Database Services

- database_services.png

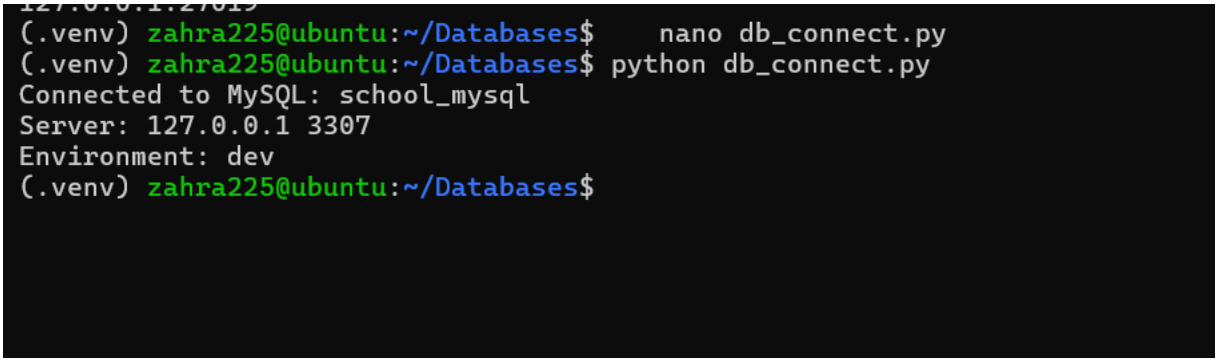


- database_ports.png



Task 4: MySQL with Values in the Code

- mysql_hardcoded_connection.png



Task 5: MongoDB with Values in the Code

- mongo_hardcoded_connection.png

```
(.venv) zahra225@ubuntu:~/Databases$  nano db_connect.py
(.venv) zahra225@ubuntu:~/Databases$    python db_connect.py
Connected to MongoDB: school_dev
Server: ('127.0.0.1', 27017)
Environment: dev
(.venv) zahra225@ubuntu:~/Databases$
```

Task 6: MongoDB with os.getenv()

- **mongo_env_dev.png**

```
Connected to MongoDB: school_dev
Server: ('127.0.0.1', 27017)
Environment: dev
(.venv) zahra225@ubuntu:~/Databases$
```

- **mongo_env_test.png**

```
Connected to MongoDB: school_test
Server: ('127.0.0.1', 27018)
Environment: test
(.venv) zahra225@ubuntu:~/Databases$
```

- **mongo_env_prod.png**

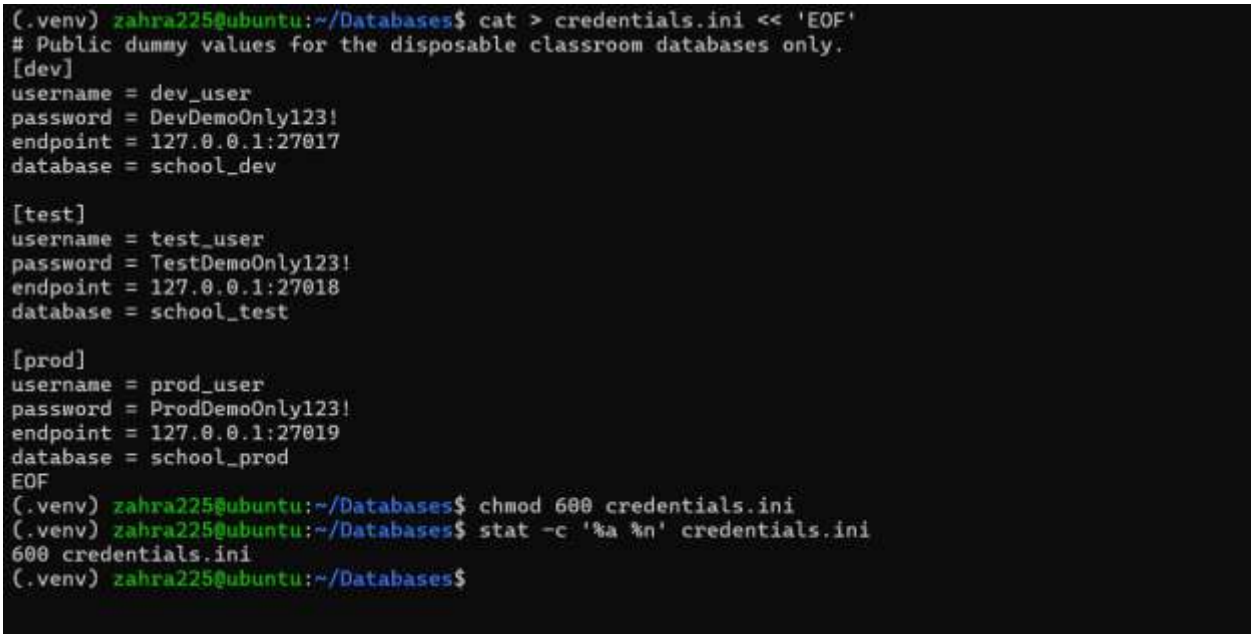
```
zahra225@ubuntu: ~/Databa
Connected to MongoDB: school_prod
Server: ('127.0.0.1', 27019)
Environment: prod
(.venv) zahra225@ubuntu:~/Databases$
```

Task 7: MongoDB with a Credentials File

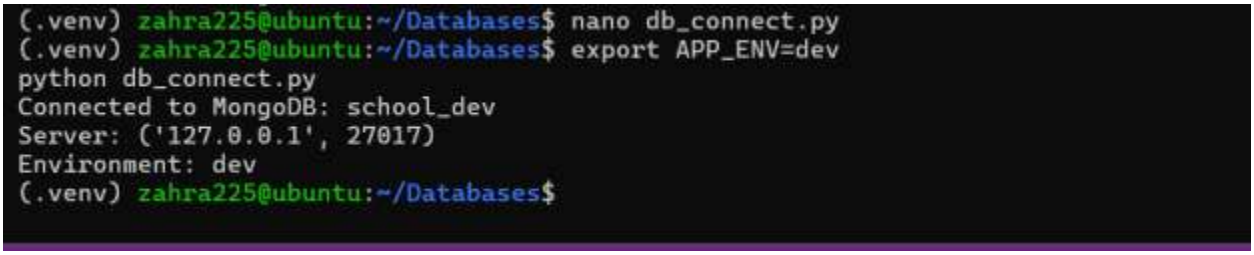
- **credential_files_created.png**



- **credential_file_permissions.png**



- **mongo_file_dev.png**



- **mongo_file_test.png**

```
(.venv) zahra225@ubuntu:~/Databases$ export APP_ENV=test
python db_connect.py
Connected to MongoDB: school_test
Server: ('127.0.0.1', 27018)
Environment: test
```

- **mongo_file_prod.png**

```
(.venv) zahra225@ubuntu:~/Databases$ export APP_ENV=prod
python db_connect.py
Connected to MongoDB: school_prod
Server: ('127.0.0.1', 27019)
Environment: prod
(.venv) zahra225@ubuntu:~/Databases$
```